MICAH BAXTER

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EDUCATION

Bachelor of Science in Mechanical Engineering Expected May 2029

Liberty University, Lynchburg, VA

- Physics Minor, 3.85 GPA
- Organizations: Liberty Rocketry Competition Team
- Full Academic Scholarship, ACT: 35

TECHNICAL SKILLS

Software & Programming: SolidWorks, Onshape, Altium, KiCad, C#, C/C++, Python

Machines: Bambu Labs P1S combo (Bambu Studio), Creality Ender 3 (Cura), Shop Tools

EXPERIENCE

Chief Systems Engineer Present

Liberty Rocketry Competition Team, Lynchburg, VA

- Leading a team of 50+ engineers to develop an advanced high-power rocket for the IREC 2027 competition
- Coordinating across eight sub teams and faculty advisors, providing technical guidance through the integration of the team's first active control system
- Responsible for the design and analysis of the airbrake's thermal management system, consisting of a custom vapor chamber to protect sensitive electronics and conduct heat into the aluminum airbrake's structure.

Information Technology Intern Summer 2026

Medical Properties Trust, Birmingham, AL

- Resolved hardware and software issues including printer connectivity, email configuration, and network problems in a corporate healthcare environment
- Configured new employee workstations for onboarding, ensuring systems met security and access standards
- Assisted IT administrators in managing Active Directory including directory searches and file access permissions for employees

Avionics Design Engineer 2025-2026

Liberty Rocketry Competition Team, Lynchburg, VA

- Developed a custom STM32-based flight computer to control critical sensor and telemetry operations for a high-power competition rocket
- Appointed as Liberty Rocketry Chief Engineer for the upcoming competition season

PERSONAL PROJECTS

Thrust Vector Control Rocket 2023-2025

- Engineered an autonomous launchpad with launch clamps, a strongback, and flame trenches
- Designed and flew two TVC rockets across different airframe scales with a custom PCB flight computer and deployable catch points to enable future catch attempts, inspired by SpaceX's Starship

Embedded Systems and PCB Design 2023-2026

- Developed an STM32-based microcontroller ground computer for an upgraded launchpad that implements I2C, SPI, and UART communication and includes PWM controlled GPIO headers for modularity
- Designed and manufactured an STM32-based flight computer for a high-power collegiate rocket that gathers sensor and GPS data and transmits live telemetry for visualization on a custom ground station GUI

Puzzle Boxes 2025

- Invented and branded Baxter Boxes, a unique line of custom 3D printed puzzle boxes including HEX and a special Easter egg box with intricate details that function as mechanical moving parts